AER320 Term Test 1 formula sheet (HS ch. 1, 2, 4.1 to 4.9)

1. Units and constants

$g = 9.81$ m/s^2 $\quad W = mg \quad G = 6.673 \times 10^{-11}$ m^3/(kg·s^2)
$F = G m_1 m_2 / r^2 \quad \vec{F} = m\vec{a}$
1 lb = 4.448 N $\quad$ 1 ft = 0.3048 m $\quad$ 1 in = 25.4 mm
1 slug = 14.59 kg $\quad$ 1 kip = 1000 lb $\quad$ 1 lb·ft = 1.356 N·m
Prefixes: G 10^9, M 10^6, k 10^3, m 10^{-3}, μ 10^{-6}
Keep 4 s.f. through working, answer to 3 or 4.

2. Trig for force triangles

Law of sines: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$
Law of cosines: $c^2 = a^2 + b^2 - 2ab\cos C$
Parallelogram law: $\vec{F}_R = \vec{F}_1 + \vec{F}_2$; draw the triangle tip to tail, the angle *inside the triangle* is 180° minus the angle between the forces.
3-4-5 slope: components are $\frac{4}{5}F$ and $\frac{3}{5}F$ (check which side is which).
Resolving onto non-perpendicular axes u, v: use the triangle and the law of sines, *not* $F\cos\theta$.
Minimum force problems: the unknown force is smallest when it is perpendicular to the known resultant line.

3. Coplanar forces (2D)

$F_x = F\cos\theta, \quad F_y = F\sin\theta$ (θ from $+x$)
$F_{Rx} = \sum F_x, \quad F_{Ry} = \sum F_y$
$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}, \quad \theta = \tan^{-1}\left|F_{Ry}/F_{Rx}\right|$
then place in the right quadrant.

4. Cartesian vectors (3D)

$\vec{F} = F_x\hat{\imath} + F_y\hat{\jmath} + F_z\hat{k}, \quad F = \sqrt{F_x^2 + F_y^2 + F_z^2}$
$\hat{u}_F = \vec{F}/F = \cos\alpha\,\hat{\imath} + \cos\beta\,\hat{\jmath} + \cos\gamma\,\hat{k}$
$\cos\alpha = F_x/F$, etc.
$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
Two given angles: third from the identity; pick the sign from the figure.
Azimuth θ (in xy plane from x) and ϕ (down from $+z$):
$F_z = F\cos\phi, \; F' = F\sin\phi, \; F_x = F'\cos\theta, \; F_y = F'\sin\theta$
Elevation angle from the xy plane: swap sin and cos of ϕ.

5. Position vector, force along a line

$\vec{r}_{AB} = \vec{r}_B - \vec{r}_A = (x_B - x_A)\hat{\imath} + (y_B - y_A)\hat{\jmath} + (z_B - z_A)\hat{k}$
(*tip minus tail*; the force points from A *toward* B)
$\vec{F} = F\,\hat{u}_{AB} = F\,\dfrac{\vec{r}_{AB}}{|\vec{r}_{AB}|}$
Cable pulls *away from* the body it acts on.

6. Dot product

$\vec{A}\cdot\vec{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$
$\hat{\imath}\cdot\hat{\imath} = 1, \; \hat{\imath}\cdot\hat{\jmath} = 0$
Angle: $\theta = \cos^{-1}\left(\dfrac{\vec{A}\cdot\vec{B}}{AB}\right)$, $0 \le \theta \le 180°$
Parallel component (projection): $F_\parallel = \vec{F}\cdot\hat{u}_a$
vector form $\vec{F}_\parallel = (\vec{F}\cdot\hat{u}_a)\,\hat{u}_a$
Perpendicular component: $F_\perp = \sqrt{F^2 - F_\parallel^2}$
or $\vec{F}_\perp = \vec{F} - \vec{F}_\parallel$
Negative projection means the component points opposite $\hat{u}_a$.

7. Cross product

$\vec{C} = \vec{A}\times\vec{B}$, $C = AB\sin\theta$, direction by right-hand rule
$\hat{\imath}\times\hat{\jmath} = \hat{k}, \; \hat{\jmath}\times\hat{k} = \hat{\imath}, \; \hat{k}\times\hat{\imath} = \hat{\jmath}$ (reverse order gives $-$), $\hat{\imath}\times\hat{\imath} = \vec{0}$
$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$
$= (A_yB_z - A_zB_y)\hat{\imath} - (A_xB_z - A_zB_x)\hat{\jmath} + (A_xB_y - A_yB_x)\hat{k}$
Watch the minus on the $\hat{\jmath}$ term.
$\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$.

8. Moment of a force

Scalar: $M_O = F\,d$, d = perpendicular distance from O to the line of action.
Sign: CCW + (right-hand rule, $+\hat{k}$ out of page).
Varignon / components (2D): $M_O = xF_y - yF_x$
(x, y = any point on the line of action, relative to O)
Vector: $\vec{M}_O = \vec{r}\times\vec{F}$, $\vec{r}$ from O to *any* point on the line of action.
Resultant: $\vec{M}_{R_O} = \sum(\vec{r}\times\vec{F})$
Force through O, or $\vec{r} \parallel \vec{F}$: zero moment.
Largest moment for given $|\vec{F}|$ at a point: $\vec{F}\perp\vec{r}$, $M_{max} = F\,|\vec{r}|$.
Minimum force for a required moment at a point: $F_{min} = M/|\vec{r}|$, $\vec{F}\perp\vec{r}$.

9. Moment about an axis a

$M_a = \hat{u}_a\cdot(\vec{r}\times\vec{F}) = \begin{vmatrix} u_{a_x} & u_{a_y} & u_{a_z} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$
$\vec{r}$: from *any point on the axis* to *any point on the line of action*.
Vector: $\vec{M}_a = M_a\,\hat{u}_a$. Sign + means along $+\hat{u}_a$.
Shortcut: about a coordinate axis, $M_x = \hat{\imath}\cdot\vec{M}_O$ = x component of $\vec{M}_O$.
Scalar: $M_a = F\,d_a$ if $\vec{F}\perp$ axis. Forces parallel to, or intersecting, the axis give $M_a = 0$.

10. Couples

Two equal, opposite, non-collinear forces.
$\sum\vec{F} = \vec{0}$.
Scalar: $M = F\,d$, d = perpendicular distance between the lines of action.
Vector: $\vec{M} = \vec{r}\times\vec{F}$, $\vec{r}$ from any point on $-\vec{F}$ to any point on $\vec{F}$.
Free vector: same effect anywhere on the body.
Resultant $\vec{M}_R = \sum\vec{M}$.
Couple from two points a distance d_0 apart with the forces at angle φ to the joining line: $M = Fd_0\sin\varphi$.
Equivalent couples: same magnitude and direction of $\vec{M}$.

11. Moving a force off its line

Sliding along its line: no change (transmissibility).
Moving $\vec{F}$ from A to O: add couple $\vec{M} = \vec{r}_{OA}\times\vec{F}$.

12. Equivalent force-couple system at O

$\vec{F}_R = \sum\vec{F}, \qquad \vec{M}_{R_O} = \sum\vec{M}_C + \sum\vec{r}\times\vec{F}$
2D: $F_{R_x} = \sum F_x, \; F_{R_y} = \sum F_y, \; M_{R_O} = \sum M_C + \sum M_O$
$\vec{F}_R$ is the same at every point; $\vec{M}_{R_O}$ depends on O.
Transfer: $\vec{M}_{R_P} = \vec{M}_{R_O} + \vec{r}_{PO}\times\vec{F}_R$

13. Reduce to a single force

Possible when $\vec{F}_R \perp \vec{M}_{R_O}$: always for concurrent, coplanar or parallel systems.
Distance: $d = M_{R_O}/F_R$; put $\vec{F}_R$ on the side that reproduces the sense of M_{R_O}.
2D, where it crosses a horizontal beam: $x = M_{R_O}/F_{R_y}$
(check sign: $M_O = xF_{R_y}$ for a point on the x axis)
Vertical member: $y = -M_{R_O}/F_{R_x}$ (from $M_O = -yF_{R_x}$)
Parallel forces along z, point (x, y) in the xy plane:
$F_R = \sum F_z, \; M_{R_x} = \sum yF, \; M_{R_y} = -\sum xF$
$x = -M_{R_y}/F_R, \qquad y = M_{R_x}/F_R$
or directly: $\bar{x} = \sum xF/\sum F$, $\bar{y} = \sum yF/\sum F$

14. Wrench (general 3D reduction)

$M_\parallel = \vec{M}_{R_O}\cdot\hat{u}_{F_R} \qquad \vec{M}_\perp = \vec{M}_{R_O} - M_\parallel\,\hat{u}_{F_R}$
Wrench: $\vec{F}_R$ plus couple $M_\parallel\hat{u}_{F_R}$ on the axis through P, where $\vec{r}_{OP}\times\vec{F}_R = \vec{M}_\perp$.
$M_\parallel > 0$: positive wrench (same sense as $\vec{F}_R$); < 0: opposite.
With $\vec{r}_{OP} = (x, y, 0)$:
$\vec{r}\times\vec{F}_R = (yF_z, \; -xF_z, \; xF_y - yF_x)$

15. Distributed loads $w(x)$ [N/m]

$F_R = \int_L w\,dx$ = area under the load curve
$\bar{x} = \dfrac{\int x\,w\,dx}{\int w\,dx}$ = centroid of that area
Rectangle w_0 over L: $F = w_0L$ at $L/2$.
Triangle 0 to w_0: $F = \frac{1}{2}w_0L$ at $\frac{L}{3}$ from the *large* end ($\frac{2L}{3}$ from zero end).
Trapezoid w_1 to w_2: split into rectangle (w_1L at $L/2$) + triangle ($\frac{1}{2}(w_2 - w_1)L$ at $\frac{2L}{3}$ from the w_1 end), then combine with $\bar{x} = \sum\tilde{x}F/\sum F$.
Spandrel $w = kx^n$ ($0 \le x \le L$, peak w_0 at $x = L$): $F = \dfrac{w_0L}{n+1}$ at $\bar{x} = \dfrac{n+1}{n+2}L$.
Pressure on a plate of width b: $w = p\,b$.

16. Equilibrium (next topic, just in case)

Particle: $\sum\vec{F} = \vec{0}$
($\sum F_x = \sum F_y = \sum F_z = 0$)
Rigid body: $\sum\vec{F} = \vec{0}, \; \sum\vec{M}_O = \vec{0}$
Spring: $F = ks$, s = stretch from unstretched length
Pulley (frictionless): same tension both sides.

17. Procedure and traps

- Sketch, label axes, choose + sense for moments before writing a line.
- Position vectors: tip minus tail. Always check $|\hat{u}| = 1$.
- Couple moments add directly to $\sum M$, no r needed, any point.
- In $\vec{r}\times\vec{F}$ order matters: $\vec{r}$ first.
- Answer with magnitude *and* direction (angle + quadrant, or unit vector, or CW/CCW).
- Units on every line: N·m, kN, lb·ft.